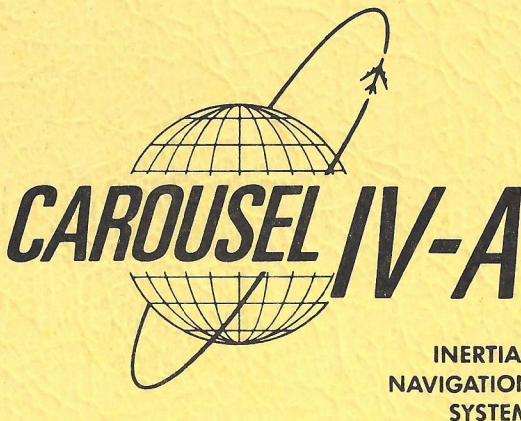


Bucher

**ON AIRPLANE
REFERENCE
HANDBOOK**



TRAINING MATERIAL ONLY

THIS MANUAL IS FURNISHED EXCLUSIVELY
FOR THE OPERATION AND MAINTENANCE OF
THE CAROUSEL IV-A INERTIAL NAVIGATION
SYSTEM, NO OTHER RIGHTS ARE GRANTED.



Delco Electronics

GENERAL MOTORS CORPORATION MILWAUKEE, WISCONSIN 53201

CAROUSEL IV-A

THIS HANDBOOK IS FOR USE AS A TRAINING AID IN OPERATING THE CAROUSEL IV-A INS. THE INFORMATION IS FOR REFERENCE ONLY AND SHOULD NOT BE USED IN PLACE OF APPROVED OPERATION AND MAINTENANCE MANUALS.

CIV INS OPERATION

A CIV Navigation Unit loaded with the CIV-A operational flight program has the following significant operational differences when compared with a CIV NU loaded with a CIV flight program:

- The CIV NU can be operated as a normal inertial navigation system using the procedures in this handbook.
- The CIV NU cannot receive DME data directly from a DME receiver but can perform DME mixing if it receives slant range information from a DME connected CIV-A INS (and PI = 4 is commanded).
- The CIV NU can perform triple mixing in a three system installation if PI = 4 is commanded. The other two systems may be either CIV or CIV-A systems.

This revision of the Training Aid includes the following Operational Flight programs:

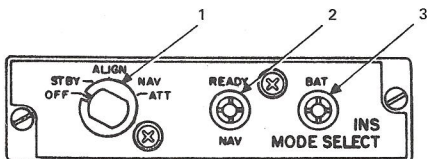
- CIV-A-19 Program, P/N 7881711-004
- CIV-AC-02 Program, P/N 7881710-004

JULY 1975

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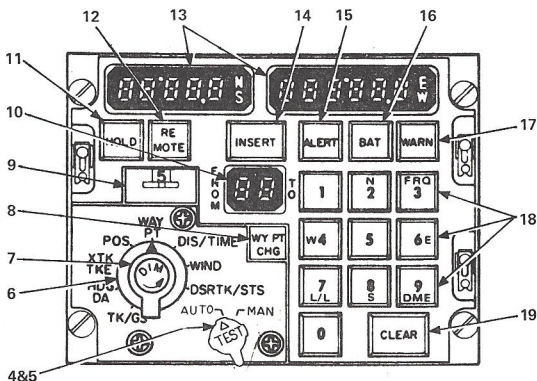
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CONTROLS AND INDICATORS



MODE SELECTOR UNIT (MSU)

1. **MODE SELECTOR SWITCH** — selects operating modes of INS: OFF STBY, ALIGN, NAV, and ATT.
2. **READY NAV INDICATOR** — lights green when INS can be advanced to NAV mode.
3. **BAT INDICATOR** — lights red if INS has shutdown while operating on battery power, thus indicating low battery charge.



CONTROL DISPLAY UNIT (CDU)

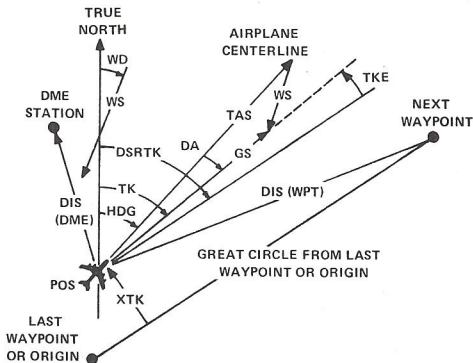
4. **AUTO/MAN SWITCH** — selects leg switching by the INS (AUTO), or the pilot (MAN).
5. **TEST SWITCH** — when pressed, causes all indicators on CDU and MSU to light and all displays to show 8's.
6. **DATA SELECTOR SWITCH** — selects which nav data is to be entered or displayed in data displays.
7. **DIM SWITCH** — controls light intensity of the data displays, the FROM-TO display, and the ALERT, HOLD, REMOTE, INSERT, and WY PT CHG indicators.

AND LAMP REPLACEMENT

8. **WY PT CHG KEY** — when lit allows numbers in FROM-TO display to be changed using keyboard.
9. **WAYPOINT/DME SELECTOR SWITCH (THUMBWHEEL SWITCH)** — selects waypoint number or DME station number for which data is to be displayed/inserted.
10. **FROM-TO DISPLAY** — two non-flashing numbers display the waypoints defining the navigation leg being used. A single flashing number defines the DME station which has been designated for mixing.
11. **HOLD KEY** — used to freeze displayed position and, in conjunction with other controls, to perform manual position update and to display inertial present position.
12. **REMOTE KEY** — when lit amber allows simultaneous loading of waypoint and DME station data into more than one INS from one CDU or one ADEU.
13. **LEFT AND RIGHT DATA DISPLAYS** — displays data from INS computer as selected by the data selector switch. Also displays data loaded by the keyboard prior to insertion.
14. **INSERT KEY** — inserts loaded data into the computer. It normally lights while data is being loaded into CDU and goes out after being pressed to insert data into the INS computer.
15. **ALERT INDICATOR** — lights amber two minutes before airplane reaches waypoint during flight if GS > 250 kts (one minute before leg switch with CIV-AC program if steering is commanded).
16. **BAT INDICATOR** — lights amber when INS is operating on battery power.
17. **WARN INDICATOR** — lights red when INS detects an out of tolerance condition, an operator error, or an error in data received from an ADEU.
18. **KEYBOARD (KEYS 0 THROUGH 9)** — consists of 10 keys to load data into data displays and FROM-TO display.
19. **CLEAR KEY** — used to cancel a data loading operation prior to insertion into the INS computer.

LAMP REPLACEMENT DATA		
INDICATOR	RATING	TYPE OR P/N
CDU BAT, WARN MSU READY NAV, BAT	28V, 40 ma	MS 18209-387 (ALT MS 25237-327)
CDU KEYBOARD KEYS & CLEAR	14V, 80 ma	CM-382
CDU HOLD, REMOTE, INSERT, ALERT, & WY PT CHG	6V, 200 ma /	MS 25237-328 8599132-001

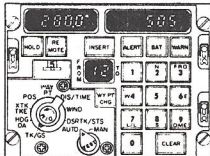
NAVIGATION DATA RELATIONSHIPS



DATA DISPLAYS

TK/GS

TRACK (LEFT DISPLAY)



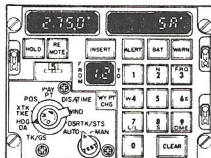
- 0 to 359.9° with respect to true north (for example, 280.0°).
- True heading is displayed instead of TK at ground speeds < 75 knots.

GROUND SPEED (RIGHT DISPLAY)

- 0 to 2,400 knots (for example, 505).
- MC31 set if > 910 knots (> 2,000 knots for supersonic operation).

HDG-DA

HEADING (LEFT DISPLAY)



- 0 to 359.9° with respect to true north (for example, 275.0°).

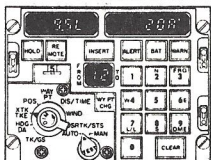
DRIFT ANGLE (RIGHT DISPLAY)

- 0 to 180° right or left of airplane heading (for example, 5 R°).
- Set to 0° if GS < 75 knots.
- MC42 set if DA > 45°.

DATA DISPLAYS

XTK-TKE

CROSS TRACK DISTANCE (LEFT DISPLAY)



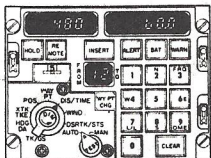
- 0 to 999.9 nmi right or left of desired track (example 9.5L).

TRACK ANGLE ERROR (RIGHT DISPLAY)

- 0 to 180° right or left of desired track angle (example, 20R°).
- Angle between desired track and heading if GS < 75 knots.

DIS-TIME

DISTANCE (LEFT DISPLAY)



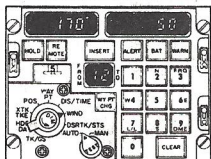
- 0 to 9999 nmi (example, 480).
- Distance to next waypoint.
- Distance between waypoints if WY PT CHG key is lit.
- Distance to DME station (see page 17 for procedure).

TIME TO GO (RIGHT DISPLAY)

- 0 to 999.9 minutes (example, 60.0).
- Time to fly distance in left display using present GS.
- Set to 999.9 if GS < 10 kts.

WIND

DIRECTION (LEFT DISPLAY)



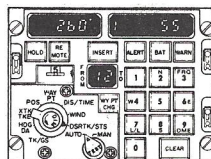
- 0 to 359° from N (example, 170°).
- Set to 0° if ADS WARN or if TAS < 152 kts or > 606 kts (>1,300 kts for supersonic operation).

SPEED (RIGHT DISPLAY)

- 0 to 606 knots (example, 50).
- Set to 0 if ADS WARN or if TAS < 152 kts or > 606 kts (>1,300 kts for supersonic operation).

DSRTK-STTS

DESIRED TRACK (LEFT DISPLAY)



- 0 to 359° (example, 260°).
- Set to 225° during STBY or ALIGN.

SYSTEM STATUS (RIGHT DISPLAY)

- See page 8 for description of STS display (example, 1 55).

INS MODES OF OPERATION

MSU CONTROL OF INS MODES OF OPERATION

- The modes of operation controlled by the MSU mode selector include STBY, ALIGN, NAV, and ATT.
- Normally only ALIGN and NAV are used. ATT is an emergency mode of operation. STBY is used if INS must be realigned or the airplane is moved prior to NAV.

ALIGN MODE NUMBER

The align mode number decreases from 9 toward 0 as alignment progresses. The INS can be advanced to NAV whenever the number is 5 or less.

No.	Mode Description
9	Standby: NU warmup at 20°F per min. Gyros run up to speed 2 minutes after warmup. Gimbals caged in pitch and roll.
8	Coarse level: Gimbals aligned to local horizontal. Battery unit is tested.
7	Coarse azimuth: Initial estimate of true north. Local latitude calculated and compared to loaded latitude.
6	Fine align: Knowledge of true north is refined. Gyros and accelerometers are calibrated (scale factors and bias).
5 ↓ 0	Refinement of alignment: Update of carousel rate. The INS may be advanced to NAV at any time during this mode with no loss of accuracy.

NAV (NAVIGATION MODE)

- NAV must be entered before moving the airplane and is the normal mode of operation during flight.
- The mode selector may be set from OFF to NAV. In this case mode 8 battery test is bypassed and NAV mode is entered as soon as mode 5 is reached.

ATT (ATTITUDE MODE)

- The attitude mode is an emergency mode used if nav data becomes unreliable or the INS computer fails.
- The CDU is blank in this mode.
- Attitude signals (pitch, roll, and platform heading) are available to other systems in this mode.
- Attitude flags on instruments are retracted during this mode, thus verifying that the attitude mode is operational.

TURN ON, ALIGN, ADVANCE TO NAV

TURN ON AND PERFORM LAMP TEST

CAUTION: ENSURE THAT COOLING AIR IS AVAILABLE TO NAV UNITS BEFORE OPERATING INS.

CAUTION: AIRPLANE MUST NOT BE MOVED DURING ALIGNMENT. IF MOVED, RECYCLE BACK THROUGH STBY.

1. Check that all applicable circuit breakers are in.
2. Check that cooling air is available to Nav Unit(s).
3. Rotate mode selector to ALIGN.
4. Perform INS lamp test (see page 8).
5. Observe CDU BAT indicator will light for 12 seconds in align mode 8 indicating battery test in process.

ENTER PRESENT POSITION

NOTE: INS will not advance beyond align mode 7 until present position is loaded.

1. Rotate data selector to POS. (Coordinates of present position at last turn off will appear in data displays).
2. Key in, confirm, and insert latitude of ramp position, starting with N/2 or S/8 to indicate north or south latitude.
3. Key in, confirm, and insert longitude of ramp position, starting with E/6 or W/4 to indicate east or west longitude.

NOTE: If CDU WARN lights immediately after insertion, check loaded position. If wrong, recycle back through STBY and reload. Clear malf (page 19).

LOAD AND VERIFY FLIGHT PLAN DATA (PAGES 9, 15, 16)

1. Verifying distance between waypoints 0 (present position) and 1 verifies present position loading.

ADVANCE TO NAV

NOTE: It is recommended that the INS be left in ALIGN until just prior to moving airplane from blocks.

1. After READY NAV indicator on MSU lights green, rotate mode selector to NAV.

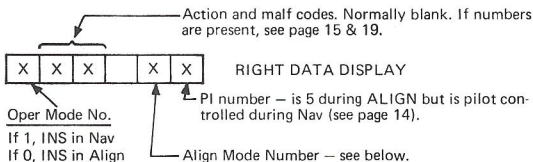
NOTE: The INS will not enter NAV until the mode number is 5 or less, even though the mode selector is at NAV.

2. Rotate data selector to DSRTK/STS and observe a 1 in first digit of right data display (see page 8). A 0 in this position indicates that the INS is still in ALIGN.

SYSTEM STATUS AND LAMP/INTERFACE TEST

SYSTEM STATUS

System status can be determined by rotating the data selector to DSRTK/STS and observing right data display.



ALIGN MODE NUMBER (ALSO SEE PAGE 6)

The mode number decreases from 9 toward 0 as alignment progresses. The INS can be advanced to NAV any time after number is 5 or less with no loss of nav accuracy.

- Number sequencing from 9 to 5 is time dependent, and is consistently the same assuming present position has been loaded.
- Number sequencing from 5 to 0 is less consistent. Under ideal conditions the decrease will be steady and smooth but wind buffeting and/or loading operations can cause a 'hang-up' at 5 for a relatively long time followed by a rather abrupt decrease to 0.

TEST INERTIAL NAV SYSTEM (LAMP TEST)

While depressing TEST switch, observe on CDU and MSU:

- 8's appear in data display and FROM-TO display.
- All indicators on CDU and MSU are lit.

INTERFACE TEST WITH OTHER A/C SYSTEMS (IN ALIGN)

1. Ensure HSI is in INS mode and coupled to INS under test. Rotate data selector to any position except DSRTK/STS and observe the following while depressing TEST switch:
 - All angles indicated at zero degrees.
 - Cross track deviation bar at 3.75 nmi (one dot) right.
 - ALERT indicator lit and INS WARN flag retracted.
 - All 8's on digital indicators.
2. Ensure autopilot is in INS mode and coupled to INS under test (in airplanes wired for INS controlled steering) and observe appropriate indications (such as control column movement, elevator and aileron movement, flight director bar, etc.) while depressing TEST switch for 15° left bank commanded with AUTO/MAN switch at AUTO and 15° right bank commanded with switch at MAN.

REMOTE FUNCTION AND WAYPOINT LOADING

REMOTE FUNCTION

More than one INS can be loaded with waypoint coordinates or DME station data from one CDU or ADEU as follows:

1. Press REMOTE key on each participating INS (both sender and receivers) so all REMOTE keys are lit.
2. Load waypoint coordinates (see below) or DME station data (see page 15) using any one CDU or ADEU for loading.
3. After data is loaded, press REMOTE keys to go out.
4. Verify that the loaded data was accepted by each receiving INS. This also verifies the loading of the sending INS.

MANUAL LOADING OF WAYPOINT COORDINATES

NOTE: See page 11 for loading waypoint data with ADEU.

1. Rotate data selector to WAY PT.
2. Set waypoint/DME selector to desired waypoint number.

NOTE: Do not load waypoint 0. If in flight, do not load waypoint numbers showing in FROM-TO display because immediate track change will occur when data is inserted.

3. Key in, confirm, and insert waypoint latitude.
4. Key in, confirm, and insert waypoint longitude.
5. Repeat steps 2, 3, and 4 until all waypoint data is loaded.

VERIFY LOADED WAYPOINT COORDINATES

Either one (or both if desired) of the following two methods may be used for verification:

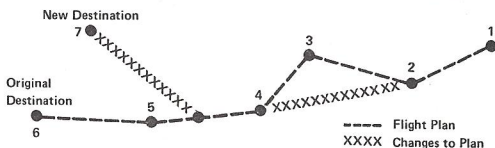
1. Check distances between successive pairs of waypoints (see below) if distances between waypoints are accurately known. (This is the preferred method).
2. Verify each loaded waypoint by monitoring latitude and longitude as waypoint/DME selector is sequentially set to each loaded waypoint number. Start the sequence by advancing one position past the last entry and rotating down in reverse direction until all entries have been verified.

DISTANCE AND TIME BETWEEN ANY TWO WAYPOINTS

1. Press WY PT CHG key until it lights.
2. Key in desired waypoint numbers in FROM-TO display. DO NOT INSERT. (Use waypoint 0 for present position).
3. Rotate data selector to DIS/TIME and read data displays.

INFLIGHT OPS – NORMAL FLIGHT

NORMAL FLIGHT PLAN



A normal flight plan includes several legs which are defined by waypoint numbers. The example flight plan shows origin as waypoint 1 and destination as waypoint 6.

FROM-TO DISPLAY

The FROM-TO display shows two waypoint numbers which define the navigation leg for which data is being calculated. For example, 1 2 means the airplane is flying from waypoint 1 to waypoint 2.

- If AUTO/MAN switch is at AUTO, FROM-TO display automatically changes to next leg when TO waypoint is reached.
- If AUTO/MAN switch is at MAN, FROM-TO display does not change when TO waypoint is reached. The pilot must manually insert the next leg numbers.

ALERT INDICATOR

The ALERT indicator lights amber two minutes before reaching TO waypoint (no ALERT if $GS < 250$ knots – one minute before leg switching with CIV-AC program if steering is commanded).

- If AUTO/MAN switch is at AUTO, ALERT goes out when TO waypoint is reached.
- If AUTO/MAN switch is at MAN, ALERT flashes on and off when TO waypoint is reached and continues flashing until a new leg is inserted in FROM-TO display or AUTO/MAN switch is rotated to AUTO.

LEG SWITCH

If AUTO/MAN switch is at AUTO when approaching TO waypoint and the autopilot is coupled to the INS, when the airplane nears the waypoint, it will automatically change track, the new leg will be displayed in FROM-TO display, and ALERT will go out.

If AUTO/MAN switch is at MAN, ALERT flashes on and off when airplane has passed TO waypoint. INS keeps on computing data for the completed leg.

IN-FLIGHT OPS – FLIGHT PLAN CHANGES

LEG SWITCH CRITERIA

If AUTO/MAN switch is at AUTO, leg switch will occur:

- For 747 aircraft at a calculated time dependent on ground speed and intercept angle.
- For other airplanes with INS controlled roll steering, about 41 seconds before reaching TO waypoint (with CIV-AC program, at a time dependent on ground speed but limited to < 120 seconds).

NOTE: *The airplane must fly each leg for at least 26 seconds before the next leg switch can occur.*

LOAD NEW WAYPOINT COORDINATES

NOTE: *Do not load data for waypoints in FROM-TO display or an immediate course change will occur.*

1. Load waypoint coordinates as described on page 9.
2. Verify inserted coordinates as described on page 9.

BYPASS WAYPOINTS

One or more waypoints may be bypassed during flight. For example, airplane has reached waypoint 2 and it is desired to bypass waypoint 3 and go to waypoint 4.

1. Press WY PT CHG key to light.
2. Load 2 4 into FROM-TO display and press INSERT. Track change occurs when INSERT is pressed.
3. Press WY PT CHG key to go out.

FLY DIRECTLY TO NEW WAYPOINT (OR DESTINATION)

A change to fly directly to a new waypoint from airplane's present position may be made at any time. For example, airplane is flying between 4 and 5 and it is desired to fly to 7.

1. Load data for waypoint 7 if not already loaded.
2. Press WY PT CHG key to light.
3. Load 0 7 into FROM-TO display and press INSERT. Track change occurs when INSERT is pressed.
4. Press WY PT CHG key to go out.

RECALL PRESENT POSITION AT TIME OF LEG CHANGE

The exact position at which a leg change was made can be displayed as follows:

1. Rotate waypoint/DME selector to 0.
2. Rotate data selector to WAY PT and read coordinates at time of leg change in data displays.

PRESENT POSITION CHECK/UPDATE

PRESENT POSITION CHECK

This check is made by "freezing" the displays at the exact instant of overflying a known reference point. Each INS must be checked individually.

1. At the exact instant when overflying a known reference point, press HOLD key to light, thus freezing the displays.
2. Rotate the data selector to POS.
3. Compare the frozen displays to reference point coordinates.
4. If both sets of coordinates are close and an update is not required, press HOLD to extinguish. This returns the displays to normal operation.

PRESENT POSITION UPDATE

After a present position check has been made (and HOLD is still lit), the calculated present position can be updated if $PI \neq 1$ — see page 14.

1. With data selector at POS, key in, confirm, and insert coordinates of the reference point.
2. If HOLD goes out and the displays start changing, the INS has accepted the update and no further action is required. The displays will show updated position.
3. If HOLD remains on and CDU WARN lights (MC 49), the INS has rejected the update as unreasonable. Do one of the following after verifying (and reloading if required) the reference point coordinates.
 - Force update by pressing HOLD to extinguish. Clear CDU Warn indication (see page 19).

NOTE: *The INS limits amount of update so a large update may be only partially accepted.*

- Purge the update by rotating data selector to DSRTK/STS and key in and insert 000001.

NOTE: *This purges all previous updates whether made manually or with DME mixing, and does not allow any new updates (sets $PI = 1$, see page 14).*

INERTIAL POSITION DISPLAY

After an update has been made, uncorrected present position (inertial present position) can be displayed.

1. Rotate data selector to WAY PT.
2. Press HOLD until it lights. The displays will show inertial present position at time update was made. Press HOLD to go out to return to previous displays.

AFTER LANDING OPERATIONS

CAUTIONS TO BE OBSERVED ON GROUND

IF GROUND POWER IS LOST BEFORE INS SHUTDOWN, THE INS WILL OPERATE ON BATTERY POWER WITHOUT COOLING AIR. THEREFORE SHUT DOWN INS IF GROUND POWER IS OFF FOR MORE THAN 2 MINUTES.

THE INS WILL NOT SHUT DOWN AUTOMATICALLY IF THE NU OVERHEATS WHILE THE MODE SELECTOR IS AT NAV. CONSEQUENTLY A CREW MEMBER SHOULD MONITOR THE INS AFTER LANDING IF THE MODE SELECTOR IS LEFT AT NAV.

EN ROUTE STOPS

NOTE: *The following are recommendations and should not be construed as requirements.*

1. If time and workload permit, (an alignment from a hot start takes about 10 minutes), downmode momentarily to STBY and then realign as described on page 7.
2. If the INS must be left unattended, rotate mode selector to ALIGN. The INS will remain in NAV but will have automatic shutdown protection if NU overheats.
3. If the INS is left in NAV and there is low residual ground speed and a large radial error, perform a present position update (see page 12).

DESTINATION

NOTE: *The following are recommendations and should not be construed as requirements.*

1. Perform a ramp radial error check (see below). Have NU pulled for poor accuracy if ramp radial error exceeds $3 + 3T$ nmi where T is navigation time in hours.
2. Rotate mode selector to OFF.

RAMP RADIAL ERROR CHECK

1. Purge all updates by inserting $PI = 1$ (see page 14).
2. Load ramp coordinates into waypoint N (N is any convenient number).
3. Press WY PT CHG key to light.
4. Key in, but DO NOT INSERT, 0 N into FROM-TO display.
5. Rotate data selector to DIS/TIME and read ramp radial error in left data display. Divide by time in NAV (in hours) to get error rate. Press CLEAR key.

CIV-A: OPERATIONAL DESCRIPTION

FEATURES UNIQUE TO CIV-A INS

Each CIV-A INS

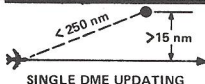
- accepts and uses slant range data from an ARINC 568 (or modified ARINC 521) Distance Measuring Equipment (DME) to update INS calculated present position.
- accepts waypoint and DME station data from an Automatic Data Entry Unit (ADEU).
- re-transmits received DME slant range data and ADEU loaded data to other connected INS systems.

CIV-A NAVIGATION MODE OPTIONS

The pilot can select three different operating modes in NAV by rotating the data selector to DSRTK/STS and keying in and inserting different PI numbers.

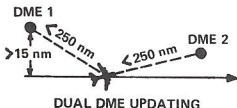
- PI = 5, 3, 2, or 0: The INS has normal inertial operation with all previous updates being retained.
- PI = 1: The INS has normal inertial operation but all previous updates are purged (not used) and no further updates can be made until the PI number is changed.
- PI = 4: The INS has aided inertial operation where slant range data from DME station(s) or triple system mixing is used to update INS calculated position.

DME UPDATING AND STATION SELECTION



Each CIV-A INS can perform single DME mixing or dual DME mixing in a multiple INS installation. The following are needed for favorable geometry:

- Slant range < 250 nmi.
- At least one station must be off desired track > 15 nmi.
- For optimum updating, DME mixing should continue until airplane has passed station.



TRIPLE INS MIXING

If three systems are operating in an aided inertial mode (that is, PI = 4) and DME updating is not taking place, a triple mixing process occurs which results in an optimum calculated present position based on the individual inertial positions. No action is required by the pilot to get this triple mixing. All three systems will display the same present position and furnish the same nav data during three system mixing.

CIV-A: DME STATION LOADING

MANUAL INSERTION OF DME STATION DATA

NOTE: The REMOTE function (see page 9) may be used for loading DME station data.

1. Rotate data selector to WAY PT.
2. Set waypoint/DME selector to desired station number.
3. Simultaneously press keyboard keys 7/LL and 9/DME.

NOTE: FROM side of FROM-TO display goes blank while a number (either last station selected or 0 if no station is selected) flashes on and off in the TO side.

4. Key in, confirm, and insert DME station latitude.
5. Key in, confirm, and insert DME station longitude.
6. Simultaneously press keyboard keys 3/FREQ and 9/DME.
7. Key in, confirm, and insert DME station altitude.

NOTE: DME station altitude is keyed in by first pressing the N/2 key followed by station altitude in thousands of feet. Only numbers 0 through 15 (corresponding to 0 to 15,000 feet respectively) are accepted by the INS.

8. Key in, confirm, and insert DME station frequency.

NOTE: Station frequency is keyed in by first pressing key 6/E followed by station frequency in MHz to the nearest 0.05 MHz. Only frequencies from 108.00 to 135.95 are accepted by the INS.

9. Repeat steps 2 through 8 until all station data is loaded.

VERIFICATION OF DME STATION DATA LOADING

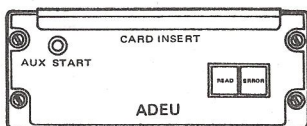
After DME station data is loaded, it can be verified by either (or both if desired) of the following methods:

1. Verify distance to DME station prior to use as described on page 17 "Verify Distance to DME Station."
2. Verify each loaded DME station by monitoring latitude and longitude as waypoint/DME selector is sequentially set to each loaded station number. Start sequencing by advancing one position past the entry and rotating down in reverse direction until all entries have been verified.

RETURNING CDU DISPLAYS TO NORMAL OPERATION

To clear CDU displays of DME information, rotate data selector to some position other than WAY PT or DIS/TIME. This will not affect DME mixing.

CIV-A ADEU OPERATION



AUTOMATIC DATA ENTRY UNIT (ADEU)

LOADING FLIGHT PLAN DATA WITH ADEU

NOTE: *More than one INS can be loaded with one ADEU using the REMOTE function (see page 9).*

1. Insert card into CARD INSERT slot of ADEU with side 1 (coded side) of card up and heavy black line on the card to the left. The ADEU will automatically pull in card and will eject the card after reading it.
2. The READ indicator lights green while the card is being read.
3. The ERROR indicator lights red if the ADEU detects a card read error.
4. The AUX START button is used to manually restart the ADEU if the card jams inside the ADEU.

INS ACCEPTANCE OF CARD DATA FROM ADEU

The card can contain information for up to seven waypoints and nine nav-aids. The INS stores this data as follows:

1. The first waypoint read is stored in the waypoint location one higher than the number in TO portion of the FROM-TO display and loading continues to the location one lower than FROM waypoint number. Waypoint 0 is not loaded. CARE MUST BE TAKEN THAT THE FROM-TO DISPLAY SHOWS THE APPROPRIATE LEG WHEN LOADING WAYPOINT DATA.
2. Each DME data group on the card has an assigned number dependent on its position on the card. The INS stores each group by this assigned number.
3. If the INS detects an error in card data, it issues action code 05 with malf code 55 and lights CDU WARN. The INS being loaded from the ADEU will not transmit this data to the other INS if it detects an error even though the REMOTE function is being used.
4. The data should be verified after loading is completed.

CIV-A: OBTAINING AIDED OPERATION

SELECTING AIDED INERTIAL MODE

NOTE: Perform for each INS individually.

1. Set data selector to DSRTK/STS.
2. Key in the desired PI number (4 for aided inertial), confirm number, and insert.

VERIFY DISTANCE TO DME STATION

NOTE: Perform for each INS connected to a DME receiver.

1. Tune DME receiver to selected DME station.
2. Set waypoint/DME selector to number of selected station.
3. Set data selector to DIS/TIME.
4. Simultaneously press keyboard keys 7/LL and 9/DME.

NOTE: FROM side to FROM-TO display goes blank while a number (either last station selected or 0 if no station is selected) flashes on and off in the TO side.

5. Check that distance to DME station in left data display approximates the DME distance displayed on the HSI.

DESIGNATE DME STATION TO BE USED FOR UPDATE

NOTE: Perform for each INS connected to a DME receiver after verifying distance to station.

1. Press WY PT CHG key until it lights. INSERT key lights and TO side stops flashing.
2. Key in, confirm, and insert number of desired DME station. WY PT CHG and INSERT go out and TO side starts flashing.

NOTE: A DME MIXING discrete (which lights a panel annunciator) is normally issued within 12 seconds after station designation.

NOTE: A flashing 0 in the TO display about 2.5 minutes after station designation indicates that DME data has failed reasonableness tests.

SWITCHING FROM AIDED TO UNAIDED OPERATION

- Recommended method — Designate 0 as the DME station (see above) number which will maintain PI = 4 thus retaining three system mixing.
- Alternate No. 1 — Insert PI = 5, which will retain previous updates but will lose three system mixing.
- Alternate No. 2 — Insert PI = 1 which will purge all updates, with no further updates being possible until a new PI number is loaded.

PILOT RESPONSE TO COMMON INS PROBLEMS

INS DEAD AT TURN ON (CDU & MSU BLANK)

The INS requires both battery power and 400 Hz for turn on.

1. Check if battery is installed and its circuit breaker is in.
2. Check that INS 400 Hz circuit breakers are in.

INS SHUTS DOWN DURING OPERATION (CDU WARN ON)

1. If MSU BAT indicator is off, shutdown was due to NU overtemp. Check airplane INS cooling system. Replace NU at earliest opportunity.
2. If MSU BAT indicator is lit, shutdown was due to low battery voltage. Check that 400 Hz power is available to INS. If 400 Hz is available, go to next step.

INS TURN ON WITH LOW BATTERY POWER

If the INS battery fails the battery test during ALIGN, test can be bypassed as follows (if MEL permits):

1. Turn mode selector to OFF and then to NAV.
2. Enter present position and flight plan data as usual.
3. Rotate data selector to DSRTK/STS and monitor right data display. If alignment below mode 5 is desired, rotate mode selector back to ALIGN when second digit from left changes from 8 to 7. If the mode selector is left at NAV, the INS will enter NAV when the second digit reaches 5.

INS ATTITUDE FLAGS – NO CDU WARN

These flags are set if the INS detects loss of synchro excitation or has detected a synchro output failure:

1. Decouple all instruments using INS attitude data.
2. Ensure that 26 vac circuit breakers are closed.
3. The INS can be used for navigation data but the NU should be replaced at earliest opportunity if excitation is good.

CDU DISPLAYS LOCKUP DURING LOADING OPERATION

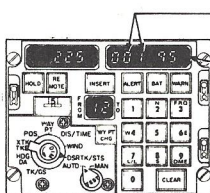
Problem may be due to temporary hangup of a numbered key. Rotate data selector to DSRTK/STS and press and release number key corresponding to rightmost digit in right data display. Try to reload.

CDU DISPLAY FROZEN OR BLANK – NO CDU WARN

Problem most likely a CDU failure. The INS may be used for attitude data but the CDU should be replaced at the earliest opportunity.

PILOT RESPONSE TO CDU WARN

ACTION AND MALFUNCTION CODES



Recommended Action or
Malfunction Code

Right Data Display
X Y Y - X X

X = Don't Care
YY = Action Code if 01 thru 05
= Malf Code if 13 thru 63

CDU WARN INDICATOR IS LIT AND DISPLAYS ARE NORMAL

CAUTION: DURING FLIGHT, SWITCH AUTOPILOT/FLIGHT DIRECTOR AND INSTRUMENTS TO A STANDBY SYSTEM IF THEY ARE COUPLED TO MALFUNCTIONING INS.

1. Rotate data selector to DSRTK/STS.
2. Observe and record action code (a number from 01 through 05) in second and third digit of right data display.
3. Press and release TEST switch.
4. Observe and record malfunction code (a number from 13 through 63) in second and third digits.
5. Repeat steps 3 and 4 until second and third digits either repeat an action code or go blank.
6. If the two digits go blank, the failure was intermittent and has been cleared and the INS can be used normally.
7. If the action code repeats, perform the recommended action shown below.

CODE	RECOMMENDED ACTION
01	Shut down INS.
02	Watch for degradation. Select autopilot coupled to other INS.
03	Check 26 V 400 Hz circuit breaker. Data to HSI/Auto-pilot unreliable.
04	(Ground only) Downmode to STBY and realign.
05	Manually load data if unable to reload via ADEU.

LINE MECHANIC RESPONSE TO CDU WARN

"RULE OF THUMB" GUIDELINES

The following guidelines are general in nature.

- Most malfunctions in the NU light CDU WARN. Conversely, failures in the CDU and MSU do NOT light CDU WARN.
- Malfunctions of the CDU are usually obvious because they result in abnormal CDU displays without lighting CDU WARN indicator.
- Malfunctions in the Battery Unit usually result in an automatic INS shutdown when the battery is tested during coarse align. In this case the CDU goes blank about 7 minutes after turn-on and MSU BAT indicator lights red.
- If a navigation instrument (ADI, HSI, autopilot) shows an instrument warning and CDU WARN is not lit, suspect the navigation instrument or associated aircraft wiring, although an NU problem is also possible.
- If there are wind display problems, suspect the air data set or associated sensors.

CDU WARN INDICATOR IS ON	
INDICATION	RECOMMENDED ACTION
CDU becomes blank after operating for some time — MSU BAT is off.	Probable automatic shutdown due to overtemp. Check NU rack cooling. Replace NU if overheating occurred.
CDU becomes blank — MSU BAT light is on.	Probable automatic shutdown due to low battery power. Replace BU for reconditioning. Check NU battery charger.
Action and malf codes are displayed when CDU data selector is in DSRTK/STS.	Action and malf codes are digits from 01 through 63 displayed in 2nd and 3rd digit of right display. See page 19.
CDU displays show erratic data.	1. Replace NU. 2. Replace CDU.
CDU displays do not change as data selector is rotated.	1. Replace NU. 2. Replace CDU.

LINE MECHANIC RESPONSE WHEN NO CDU WARN

CDU WARN INDICATOR IS OFF	
INDICATION	RECOMMENDED ACTION
INS does not turn on — CDU is blank and pressing CDU TEST switch has no effect.	1. Check battery circuit breaker. 2. Check 115 V 400 Hz circuit breaker. 3. Replace NU.
CDU is blank and WARN and BAT indicators light when TEST is pressed.	1. Verify that Mode Selector is not at ATT. 2. Check DIM control full CW. 3. Replace CDU.
One or more lamps/displays on CDU or MSU do not light when TEST is pressed.	1. Replace lamp or display. 2. Replace CDU or MSU.
CDU displays are erratic — other (HSI, ADI, etc.) nav instruments are normal.	1. Replace CDU. 2. Replace NU.
CDU INSERT key is lit.	1. Press CLEAR — if INSERT remains on enter present position — see page 7.
CDU data displays do not change when data selector is rotated and INSERT is not lit.	1. Check for abnormal switch rotation (no detent or 360° rotation possible). 2. Replace CDU.
CDU data displays blink on and off.	1. Replace CDU. 2. Replace NU.
WIND readings on CDU while on ground.	1. Air Data Set problem. 2. Replace NU.
Numerical data cannot be loaded into CDU.	1. Possible intermittent key. a. Rotate data selector to DSRTK/STS. b. Press, several times, key corresponding to right-most digit in right display. c. If still cannot load data, replace CDU.
CDU BAT indicator is lit amber (except during the battery test of mode 8).	1. Check that 115 V 400 Hz power is available. 2. Replace NU.

LINE MECHANIC RESPONSE TO MALF CODES

CODE	MALF DESCRIPTION	RECOMMENDED ACTION
01-33 01-34 01-35 01-36	Azimuth Stab Loop Inner Roll Stab Loop Pitch Stab Loop Accelerometer Loop	Replace NU if one of these loop malfs has occurred.
01-37 01-38 01-39	Z Platform Overtemp XY Platform Overtemp Oven Overtemp	1. Check NU cooling air, and 2. Replace NU.
02-32 02-59 02-63	Memory Parity Error 600 msec Loop Computer Self-Test	Replace NU if one of these computer malfs occurs (if spare NU not available and local rules permit, INS is usable as an attitude reference).
02-13 02-14 02-15 02-18 02-31 02-42 04-45 02-49 04-57 04-60 04-61 02-62	ΔY Velocity NAV ΔX Velocity - NAV X or Y Gyro Torque too high - ALIGN, NAV Too long in mode 8 Ground speed too great. Drift angle $> 45^\circ$ Change in Gyro Scale Factor or Wrong Lat was Loaded Position update > 33 nmi Gimbal angle rate too large in ALIGN ΔWSH too large WSH too large Gimbal angle rate too large in NAV	1. Align to mode 5 and enter NAV - monitor for malfs. 2. If no malfs occur, INS is good. 3. If MC 60 or 61 occur, set mode selector to STBY and then realign to mode 5. If MC 60 or 61 recur, replace NU. 4. If malfs other than 60 or 61 occur, replace NU. NOTE: If spare NU not available and local rules permit, the INS may be used as an attitude reference.
03-22* 03-23* 03-24* 03-25* 03-26*	TK Feedback DA Feedback TKE Feedback HDG Feedback XTK Feedback	1. Ensure INS is in ALIGN/NAV 2. Ensure 26 V 400 circuit breakers are in. 3. If malfs clear, INS is OK. If not, replace NU (if spare not available, INS may be used for nav data).
04-43 04-41	Loaded present position between 2 or 3 INS differ > 0.7 nmi. Loaded present position > 76 nmi from last inertial position.	1. Check loaded present position. 2. If required, downmode to STBY and reload. 3. Clear malf.
05-55	Data from ADEU fails reasonableness test.	1. If ADEU good, replace NU. 2. Data may be manually loaded.

*Malf does not set CDU WARN.

747 AUTOPILOT INTERFACE TEST

ENTER SPECIAL PRESENT POSITION

1. Set mode selector to STBY.
2. Enter 50° 00.0' N and 000° 00.0' E for present position. MC41 will occur. Clear malif.
3. Set mode selector to OFF then back to ALIGN. DO NOT INSERT PRESENT POSITION. INSERT light will remain on.

LOAD SPECIAL WAYPOINT COORDINATES

1. Load the following waypoints.

WY PT NO.	LATITUDE	LONGITUDE
1	40° 00.0' N	6° 25.7' W
2	60° 00.0' N	9° 48.0' E
3	40° 00.0' N	6° 13.4' W
4	60° 00.0' N	10° 00.0' E
5	40° 00.0' N	3° 10.0' W
6	60° 00.0' N	4° 54.6' E

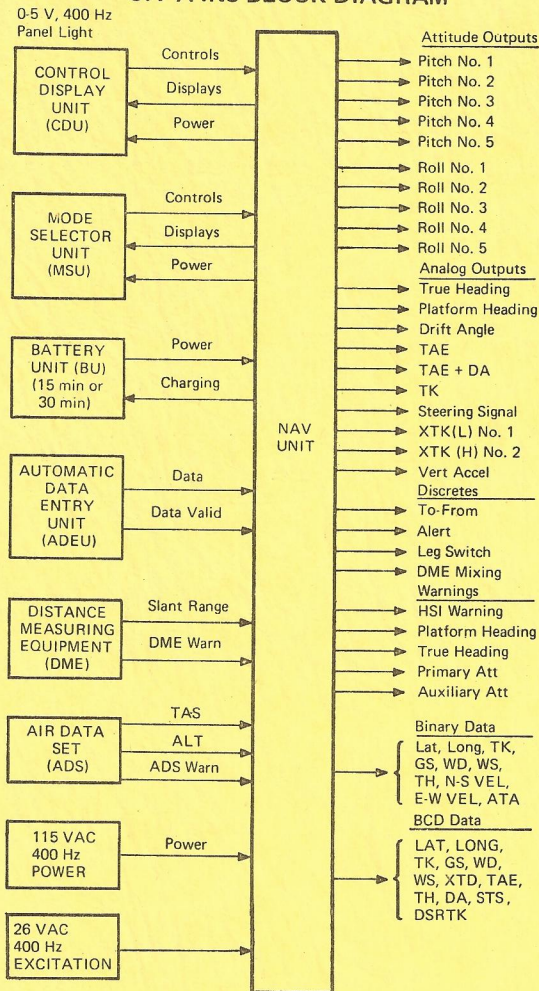
PERFORM TEST

1. Ensure INS is in mode 7 and FROM-TO is 1 2.
2. Press WY PT CHG key to light and enter, but do not insert, 00 into FROM-TO display.
3. Observe AFCS and HSI displays for values shown in the table. (Values in parenthesis can be displayed on CDU).

FROM-TO	TK	HDG	DA	XTK	TKE	WARN
1 2	22.5° (22.5°)	11.3° (11.3°)	11.25R° (11R°)	50,000R (8.2R)	15L° (15L°)	22 Vdc
3 4	22.5° (22.5°)	11.3° (11.3°)	11.25R° (11R°)	7,000R (1.1R)	15L° (15L°)	22 Vdc
5 6	22.5° (22.5°)	11.3° (11.3°)	11.25R° (11R°)	800R (0.1R)	2.4L° (3L°)	22 Vdc

4. Press WY PT CHG key to light, enter 3 4 into FROM-TO display and insert.
5. Repeat steps 2 and 3 above (for 3 4 displays).
6. Press WY PT CHG key to light, enter 5 6 into FROM-TO display and insert.
7. Repeat steps 2 and 3 above (for 5 6 displays).
8. The INS can now be returned to normal operation by entering present position or can be shut down.

CIV-A INS BLOCK DIAGRAM





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